

SECURITY LOG INVESTIGATION

Authentication Log Analysis, Suspicious Activity Detection & Incident Reconstruction

Project Type: Cybersecurity / Security Operations / Digital Forensics

Dataset: Synthetic Authentication Logs

Tools: Microsoft Excel, Python, Pandas, OpenPyXL, ReportLab, GitHub

Deliverables: Investigation Spreadsheet + Incident Report PDF

This report documents a structured investigation of synthetic authentication events to identify suspicious login behavior, detect possible brute-force and password-spraying activity, reconstruct a likely incident timeline, and recommend containment and improved logging controls.

1. Executive Summary

The investigation analyzed 15 synthetic authentication events. The records contain successful and failed login attempts involving several user accounts and source IP addresses. The activity includes repeated failed authentication attempts against the administrator account and a separate source IP targeting multiple users. These patterns are consistent with possible brute-force and password-spraying behavior.

A successful authentication for the administrator account occurs after multiple failed attempts from the same source, followed by a successful ACCESS event. This sequence warrants investigation for possible account compromise. However, the available authentication records alone do not prove malicious intent, attacker identity, credential theft, or subsequent data access.

Overall assessment: High-priority suspicious authentication activity requiring validation, containment where appropriate, and correlation with additional identity, endpoint, network, and application logs.

2. Objective

The objectives of the investigation were to:

- Normalize timestamps, IP addresses, accounts, actions, and result fields.
- Identify patterns associated with brute-force authentication attempts.
- Identify potential password-spraying activity across multiple accounts.
- Detect successful authentication following suspicious failures.
- Create a defensible incident timeline.
- Separate observed evidence from assumptions.
- Recommend containment, remediation, and improved security logging controls.

3. Evidence of Practical Implementation

The following screenshots document the practical implementation of the investigation. The first screenshot shows the normalized authentication log in Excel. The second shows the project structure and generated investigation files in Visual Studio Code. The third shows the Python investigation script executing successfully and creating the spreadsheet and incident report PDF.

PRODUCT NOTICE Most of the features of Excel have been disabled

	A	B	C	D	E	F
1	timestamp	username	source_ip	action	result	
2	#####	john	192.168.1.	LOGIN	SUCCESS	
3	#####	admin	185.10.20.	LOGIN	FAILED	
4	#####	admin	185.10.20.	LOGIN	FAILED	
5	#####	admin	185.10.20.	LOGIN	FAILED	
6	#####	admin	185.10.20.	LOGIN	FAILED	
7	#####	admin	185.10.20.	LOGIN	FAILED	
8	#####	admin	185.10.20.	LOGIN	SUCCESS	
9	#####	admin	185.10.20.	ACCESS	SUCCESS	
10	#####	alice	192.168.1.	LOGIN	SUCCESS	
11	#####	john	192.168.1.	LOGIN	SUCCESS	
12	#####	bob	185.55.22.	LOGIN	FAILED	
13	#####	alice	185.55.22.	LOGIN	FAILED	
14	#####	john	185.55.22.	LOGIN	FAILED	
15	#####	david	185.55.22.	LOGIN	FAILED	
16	#####	sarah	185.55.22.	LOGIN	FAILED	
17						
18						
19						
20						
21						

Normalized Logs Suspicious Events Incident

Ready Accessibility: Good to go

30°C Mostly cloudy

Figure 1. Normalized authentication log in Excel.

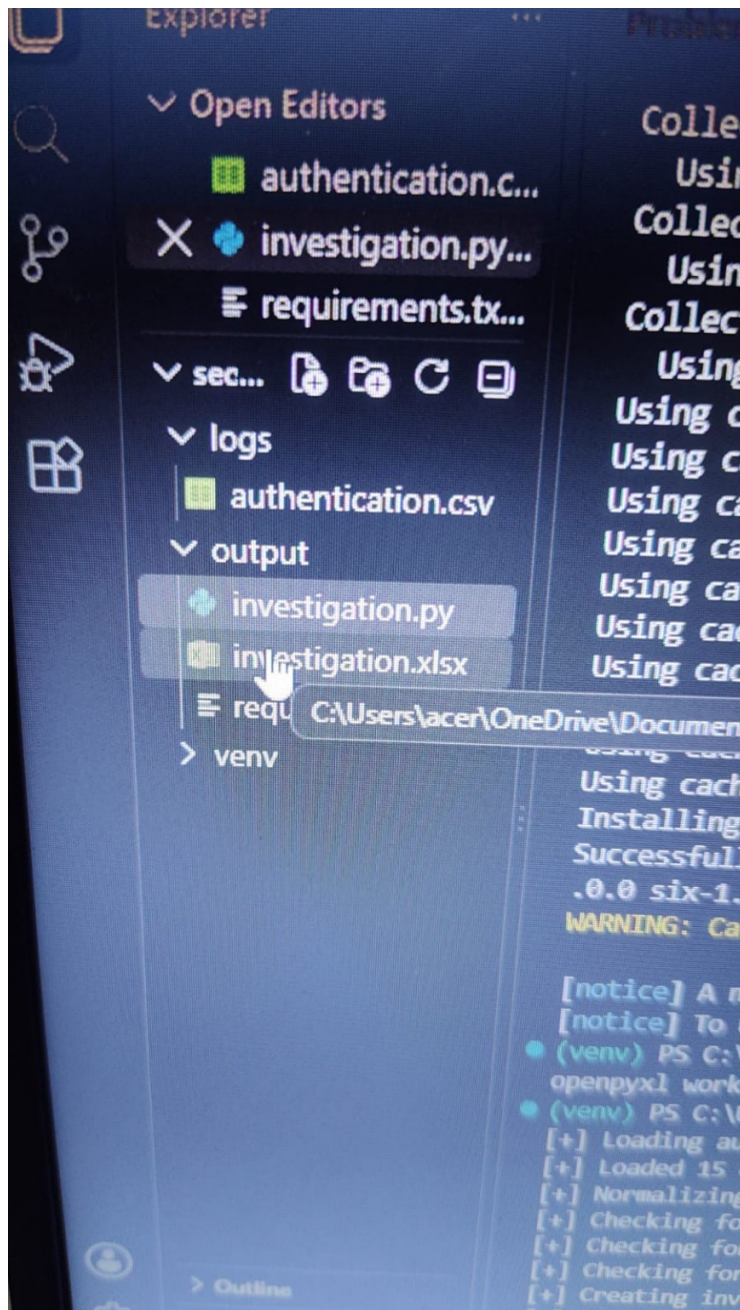


Figure 2. Project structure containing the authentication log and investigation outputs.

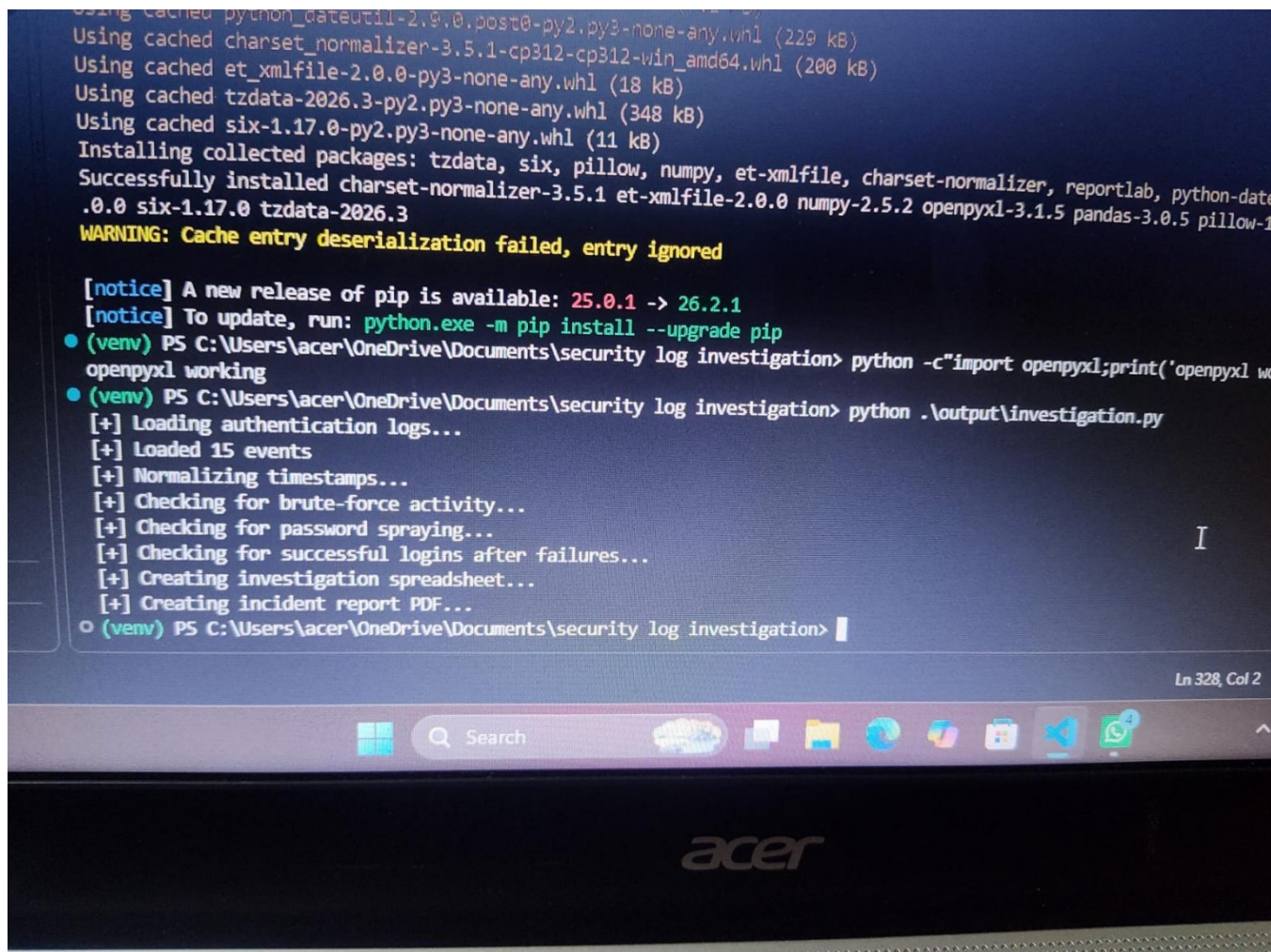


Figure 3. Successful execution of the Python investigation script and report generation.

4. Tools and Environment

Component	Purpose
Python	Automated log processing and investigation logic
Pandas	Loading, filtering, grouping, and analyzing authentication events
OpenPyXL	Creating the investigation spreadsheet
ReportLab	Generating the incident report PDF
Microsoft Excel	Reviewing normalized and suspicious events
Visual Studio Code	Developing and executing the investigation script
GitHub	Version control and project documentation

5. Dataset and Normalized Log

The practical dataset contains 15 authentication events. The visible spreadsheet shows the normalized fields timestamp, username, source_ip, action, and result. Some timestamp values are displayed as hashes in the screenshot because the Excel column is too narrow; the investigation process normalizes timestamps before analysis.

#	Username	Source IP	Action	Result
1	john	192.168.1.x	LOGIN	SUCCESS

#	Username	Source IP	Action	Result
2	admin	185.10.20.x	LOGIN	FAILED
3	admin	185.10.20.x	LOGIN	FAILED
4	admin	185.10.20.x	LOGIN	FAILED
5	admin	185.10.20.x	LOGIN	FAILED
6	admin	185.10.20.x	LOGIN	FAILED
7	admin	185.10.20.x	LOGIN	SUCCESS
8	admin	185.10.20.x	ACCESS	SUCCESS
9	alice	192.168.1.x	LOGIN	SUCCESS
10	john	192.168.1.x	LOGIN	SUCCESS
11	bob	185.55.22.x	LOGIN	FAILED
12	alice	185.55.22.x	LOGIN	FAILED
13	john	185.55.22.x	LOGIN	FAILED
14	david	185.55.22.x	LOGIN	FAILED
15	sarah	185.55.22.x	LOGIN	FAILED

Note: The screenshot visually truncates the IP addresses and timestamps due to column width. The table above preserves the visible portions without inventing hidden values.

6. Investigation Methodology

The investigation followed these stages:

1. Load the authentication CSV dataset.
2. Normalize timestamps and categorical fields.
3. Validate usernames, source IPs, actions, and results.
4. Count failed attempts by account and source IP.
5. Identify multiple-account targeting from a single source.
6. Detect successful logins following failed authentication attempts.
7. Generate suspicious-event findings.
8. Construct a chronological incident timeline.
9. Export findings to an investigation spreadsheet.
10. Generate an incident report PDF.

7. Detection Rules

7.1 Brute-Force Detection

Repeated failed authentication attempts against the same account from the same source are treated as a brute-force indicator. Thresholds should be calibrated to the organization's normal behavior.

Example logic:

```
IF failed_login_count for the same account/source exceeds a defined threshold within a short window
THEN flag as potential brute-force activity.
```

7.2 Password-Spraying Detection

Password spraying typically involves a source attempting authentication against several different accounts. A single source IP associated with failures across multiple usernames is therefore a useful detection signal.

Example logic:

IF one source IP targets multiple unique accounts AND produces repeated failures within a defined window THEN flag as potential password spraying.

7.3 Successful Login After Failures

A successful login following multiple failed attempts should receive additional scrutiny because it can represent successful credential guessing or legitimate recovery after mistyped credentials.

8. Investigation Findings

Finding 1 – Repeated Failed Attempts Against admin

The log contains five consecutive FAILED LOGIN events for the admin account from the same visible source-IP prefix (185.10.20.x), followed by a successful LOGIN and a successful ACCESS event. This is a strong indicator requiring investigation for possible brute-force activity and account compromise.

Finding 2 – Multiple Accounts Targeted by 185.55.22.x

The source-IP prefix 185.55.22.x is associated with failed LOGIN attempts against bob, alice, john, david, and sarah. This multi-account pattern is consistent with potential password spraying.

Finding 3 – Successful Authentication After Failures

The admin account successfully authenticates after multiple failures. A subsequent ACCESS SUCCESS event shows that an authenticated access action was recorded. The records do not by themselves establish whether the access was authorized or malicious.

9. Evidence vs. Assumptions

Observed Evidence	What It May Indicate	What Is Not Proven
Five failed admin logins from same source	Possible brute force	Attacker identity
Admin login succeeds afterward	Possible credential success	Credential theft
Successful admin ACCESS event	Authenticated access occurred	Malicious data access
One source targets five accounts	Possible password spraying	Intent of source
Multiple failed logins	Authentication anomaly	System compromise

10. Incident Timeline

Sequence	Observed Event	Assessment
1	Normal successful logins from 192.168.1.x	Baseline activity
2	Repeated failed admin logins from 185.10.20.x	Potential brute force
3	Admin LOGIN succeeds	Potential credential success
4	Admin ACCESS succeeds	Authenticated access recorded
5	185.55.22.x fails against bob, alice, john, david and sarah	Potential password spraying

Because the screenshot does not expose the actual timestamp values, the sequence above is based on the row order in the normalized log rather than invented clock times.

11. Risk Assessment

Risk Area	Assessment
Brute-force activity	High concern
Password spraying	High concern
Potential admin compromise	High concern
Confirmed malicious access	Not established
Data exfiltration	No evidence in supplied authentication log
Attribution	Not established

12. Recommended Containment Actions

- Validate the admin account activity with the legitimate account owner.
- Force a password reset if compromise is suspected.
- Revoke active sessions and authentication tokens where appropriate.
- Enable or enforce MFA for privileged and high-risk accounts.
- Apply login throttling and account protection controls.
- Investigate the suspicious source IPs using firewall, VPN, proxy, and threat-intelligence records.
- Review administrative changes and resource access after the successful authentication.
- Preserve relevant logs for forensic analysis before rotating or deleting them.

13. Additional Evidence Required

- Identity Provider / SSO logs
- MFA authentication logs
- Password-reset and account-lockout logs
- VPN and firewall logs
- Endpoint Detection and Response (EDR) telemetry
- Windows/Linux authentication and system logs
- Web application and API access logs
- File access and data-download records
- Cloud audit logs, if applicable
- Threat-intelligence information for the source IPs

14. Improved Logging and Detection Controls

- Centralize authentication logs in a SIEM.
- Record precise timestamps with synchronized system clocks.
- Capture username, source IP, destination application, result, authentication method, and session information.
- Log MFA success/failure and authentication risk signals.
- Create alerts for multiple failed logins against one account.

- Create alerts for one source targeting multiple accounts.
- Alert when a successful login follows an unusual sequence of failures.
- Monitor privileged accounts more closely.
- Retain logs according to organizational and legal requirements.
- Protect logs against unauthorized modification.

15. Practical Implementation Output

The supplied implementation screenshot shows the Python script being executed from a virtual environment. The console confirms that 15 events were loaded, timestamps were normalized, brute-force and password-spraying checks were performed, successful logins after failures were checked, and both the investigation spreadsheet and incident report PDF were created.

Observed execution workflow:

Loading authentication logs → Loaded 15 events → Normalizing timestamps → Checking for brute-force activity → Checking for password spraying → Checking for successful logins after failures → Creating investigation spreadsheet → Creating incident report PDF

16. Recommended GitHub Repository Structure

```
security-log-investigation/
  README.md
  logs/
  authentication.csv
  output/
  investigation.py
  investigation.xlsx
  reports/
  Security_Log_Investigation_Complete_Report.pdf
  screenshots/
  normalized-log.jpg
  project-structure.jpg
  execution-output.jpg
  requirements.txt
```

17. Learning Outcomes

- Learned how to normalize authentication log data for investigation.
- Learned to identify brute-force patterns from repeated authentication failures.
- Learned to recognize password-spraying patterns across multiple accounts.
- Learned to correlate authentication failures and subsequent successes.
- Learned to construct an evidence-based incident timeline.
- Learned to distinguish confirmed evidence from investigative assumptions.
- Learned how automation can accelerate repetitive log-analysis tasks.
- Learned the importance of centralized security logging and SIEM-based detection.

18. Conclusion

The Security Log Investigation successfully demonstrates a practical workflow for converting raw authentication records into security findings and response recommendations. The supplied dataset contains a clear cluster of failed administrator authentication attempts followed by successful authentication and access, as well as another source targeting several different accounts.

These patterns are consistent with possible brute-force and password-spraying activity and should be treated as high-priority investigation signals. At the same time, a defensible forensic report must avoid overclaiming: the supplied authentication records do not independently prove attacker identity, credential theft, data access, or exfiltration.

The practical implementation further demonstrates successful automation using Python and supporting libraries, with outputs generated as an investigation spreadsheet and incident report PDF. Recommended next steps include MFA enforcement, rate limiting, session revocation when warranted, SIEM correlation, and collection of additional identity, endpoint, network, and application evidence.

19. Final Investigation Summary

Item	Result
Dataset	15 synthetic authentication events
Primary concern	Suspicious authentication activity
Potential attacks	Brute force and password spraying
Affected privileged account	admin
Successful admin authentication	Observed
Successful admin access event	Observed
Confirmed attacker identity	Not established
Confirmed data exfiltration	Not established
Recommended priority	High – validate and contain as appropriate

End of Report

Educational cybersecurity project. Synthetic data only.